

Problem

A signal with Fourier transform $F(\omega) = \frac{20}{4 + j\omega}$ is passed through a filter whose cutoff frequency is 2 rad/s (i.e., $0 < \omega < 2$). What fraction of the energy in the input signal is contained in the output signal?

Step-by-step solution

Step 1 of 1

The total energy is

$$\begin{aligned}
 W_T &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{400}{4^2 + \omega^2} d\omega \\
 &= \frac{400}{\pi} \left[\left(\frac{1}{4} \right) + \tan^{-1} \frac{\omega}{4} \right]_{-\infty}^{\infty} = \frac{100}{\pi} = 31.831 \\
 W &= \frac{1}{2\pi} \int_0^2 \left(\frac{F(\omega)}{\omega} \right)^2 d\omega = \frac{400}{2\pi} \left[\left(\frac{1}{4} \right) + \tan^{-1} \left(\frac{\omega}{4} \right) \right]_0^2 \\
 &= \left[\frac{100}{2\pi} \right] \tan^{-1}(2) = \frac{50}{\pi} \times 1.107 = 15.814 \\
 \frac{W}{W_T} &= \frac{15.814}{31.831} = 0.4968 \text{ or } 49.68\%
 \end{aligned}$$